

LIST OF PROBLEMS: CLASSICAL AUCTION FORMATS WITH COMPLETE INFORMATION

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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1. Please look at Osborne's book chapter 3.¹ Solve exercises 82.1 (84.1), 83.1 (85.1), 84.1 (86.1), 85.1 (87.1), and 86.1 (88.1).
2. Consider the SPSA (second price sealed-bid auction) with complete information we discussed in the class. However, consider the following change: if more than one bidder submits the highest bid, each gets the object with equal probability. Formally, the tie breaking rule is now as follows: let $I = \{i \in \{1, \dots, n\} : b_i = b_{\max}(b)\}$, then the object goes with probability $1/|I|$ to any player $j \in I$. Show that it is a weakly dominant strategy for player i to submit a bid $b_i = v_i$. Is this player's i unique weakly dominant strategy?
3. Consider the FPSA (first price sealed-bid auction) with complete information that we discussed in class, and suppose there are two bidders. Assume that bids and valuations are restricted to be multiples of some discrete monetary unit ϵ (e.g. a tiyn). The tie-breaking rule is that if the two bidders submit the same bid, each gets the object with probability $1/2$.
 - (a) Are any strategies strictly dominated?
 - (b) Are any strategies weakly dominated?
 - (c) Is there a Nash Equilibrium? What is it? Is it unique?
4. Two players have a common valuation for an object given by $V > 0$. Suppose that they submit sealed bids $b_i \geq 0$, for $i = 1, 2$. However, the *probability* of winning the object by player i is now determined by the function $p_i(b_i, b_j)$ given by

$$p_i(b_i, b_j) = \begin{cases} \frac{b_i}{b_i + b_j} & b_i > 0, b_j > 0 \\ 1 & b_i > 0, b_j = 0 \\ 0 & b_i = 0, b_j > 0 \\ \frac{1}{2} & b_i = b_j = 0. \end{cases}$$

If player i submits b_i it becomes a sunk cost, i.e., she needs to pay her bid. Thus, player's i payoff is given by

$$u_i(b) = p_i(b_i, b_j) \cdot V - b_i.$$

In this exercise you will find the symmetric Nash Equilibrium of this game.

- (a) Start by maximizing player i 's payoff, assuming an *interior solution* for b_i , i.e., $b_i > 0$, and assuming that $b_j > 0$. Compute the best response functions for both players when both bids are positive.
- (b) Use the best responses to find the *unique symmetric equilibrium* of the game.
- (c) Can you generalize this exercise to n players?

¹The pdf in Moodle, which is dated November 2000, or the printed First edition, dated 2003/2004. Please note that these two versions have different numbering for the problems. When I refer to a specific problem X, I refer to the pdf version of the book in Moodle, but I will also put in parenthesis (Y) the corresponding number in the printed book.